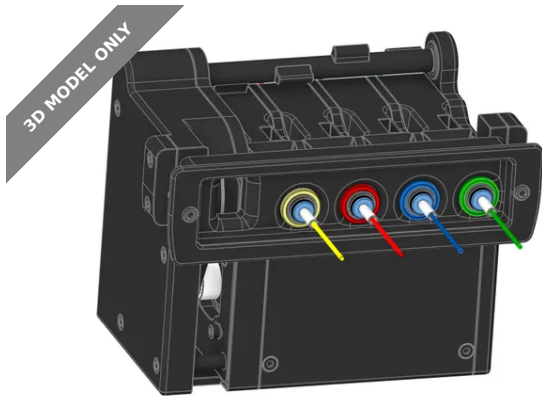




MMX Multi-Material Extruder - Exclusive Final Release



Nadir @ CN3D

[VIEW IN BROWSER](#)

updated 24. 10. 2025 | published 24. 10. 2025

Summary

Beta release of MMX! Members get 2-week exclusive access. Join our Discord for BOM & updates:

[3D Printers](#) > [Other Printer Parts & Upgrades](#)

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MMX OG — Serial Number Registration Now Open!

We're officially opening **serial number requests** for the **MMX OG (4-Color) Multi-Material Extruder**.

If you've built one and made it work — this is your moment to make it official.

Each registered unit will receive a **unique MMX Serial Number**, as recognition for your work, dedication, and for bringing this community-driven system to life.

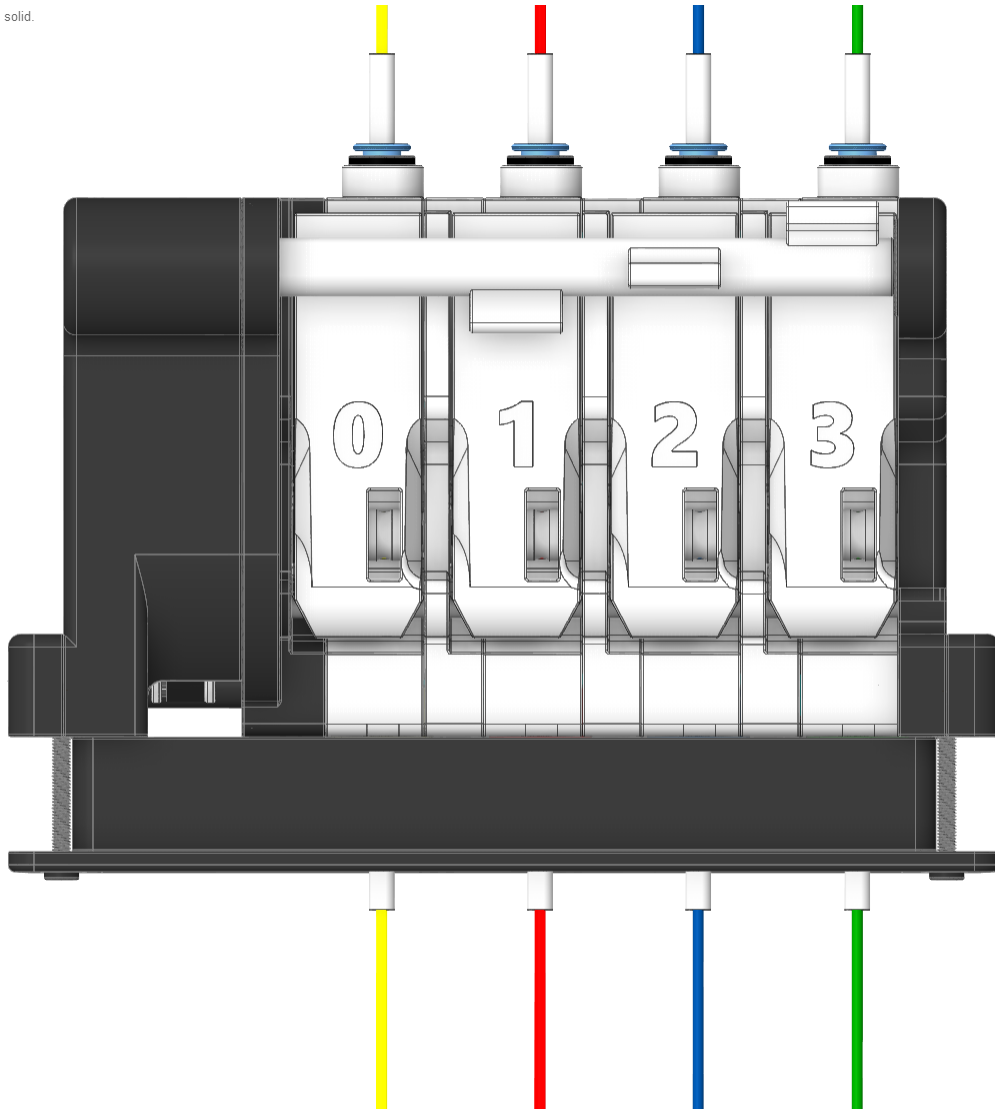
Not everyone who started the MMX journey finished it — so if you did, congratulations! You've earned your spot in the project's history.

Register your build here:

<http://cn3d.eu/mmxserialrequest>

Thank you for trusting our design and being part of the CN3D community!
Together we're showing the world how far this maker movement can go.

ck to select a solid.



We have made the idler doors a little bit narrow this way we have increased the gap where the idler door sits so it will not catch by mistake.
14-07-2025

We have updated the model now the tray has a space for a step
so if you want to have the latest model download and
print the bottom panel and the end tray
16 - June - 2025

will need: Also update the Right Side Cover.
Thanks to our member **azlaeel** for pointing this out

At the download section there is a zip file that includes Orca

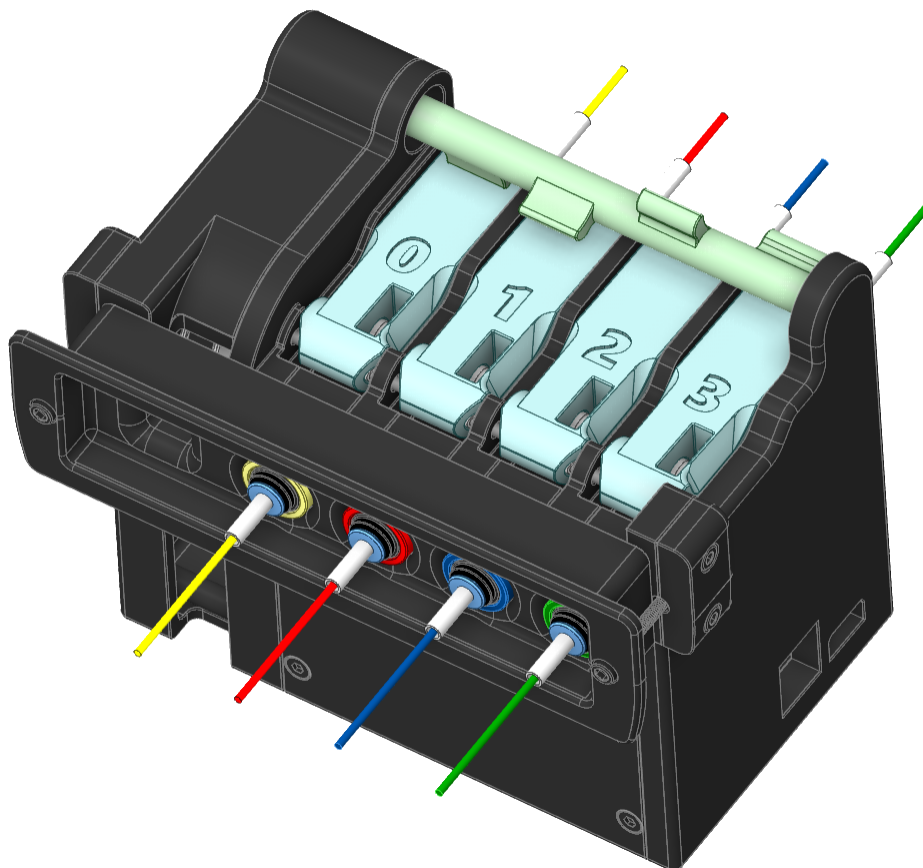
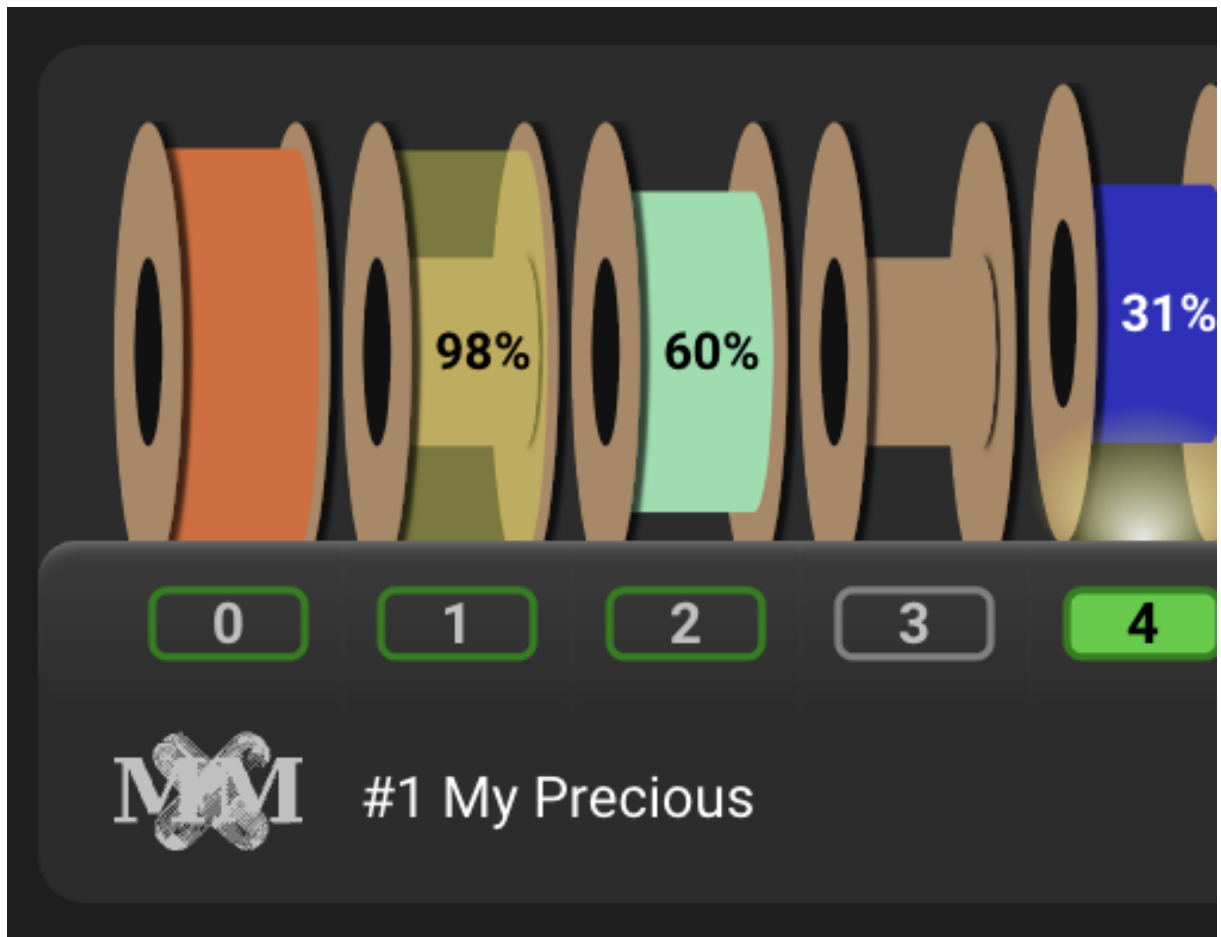
We can proudly announce that there is a kit available for purchase. The kit was tested, contains quality components and is all you need to build your MMX. You just have to download the files from here and print them.

The kit does not contain the 3d printed parts.

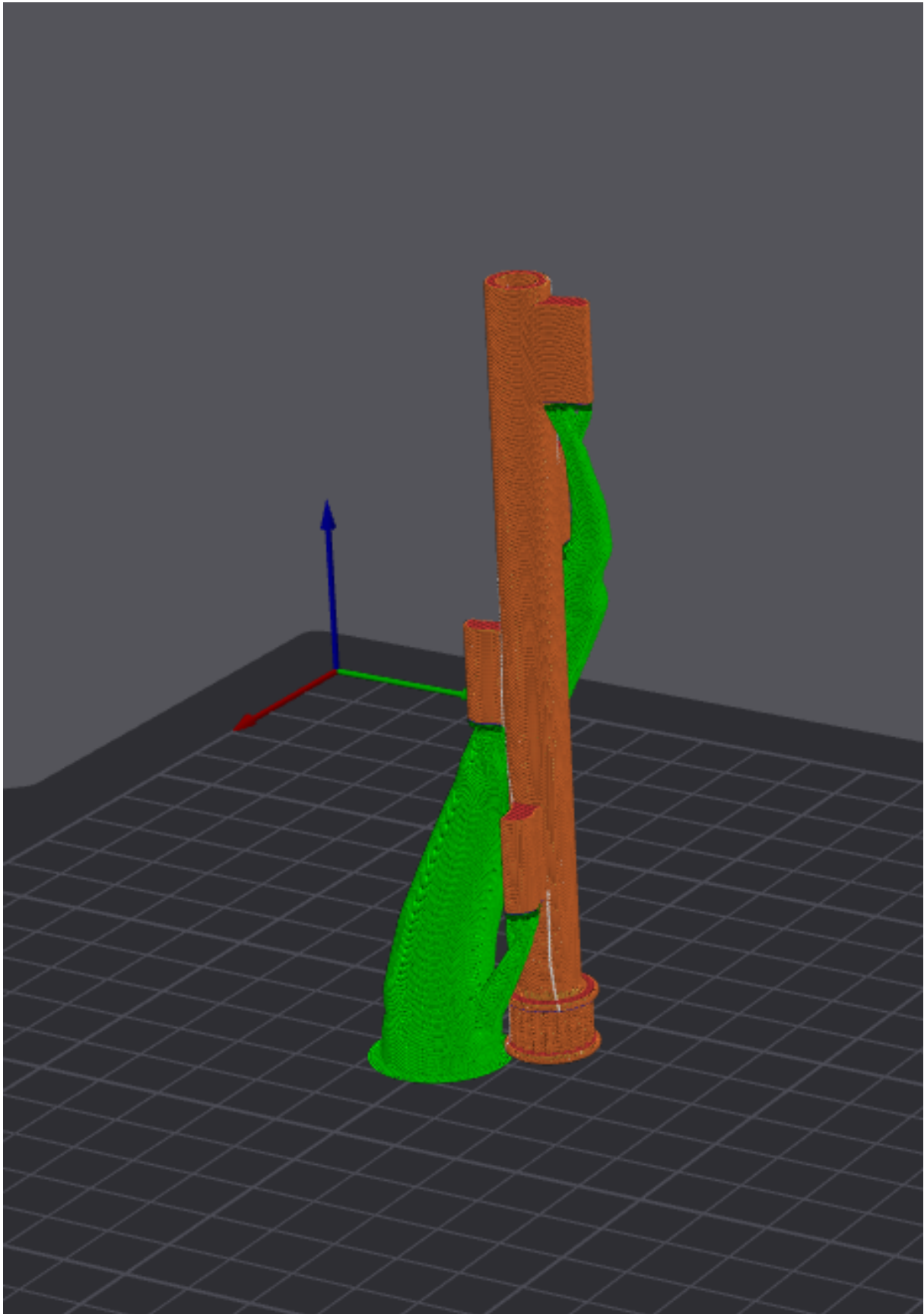
https://s.click.aliexpress.com/e/_EzjN06s

New update to the Camshaft and to the Idler doors that they are numbered to the specific locations.

also <https://github.com/moggieuk/Happy-Hare> added the MMX logo to the Mainsail PR and it's part of a new PR for Fluid



Only the camshaft need support the doors don't need



Today 3 of April I have update the design with a small change the BMG gear in the idler door sit not a little back this way we ensure the will be

engaged only when the gate is selected with the previous design it might be caught and allow the bmg to spin even when it was not selected the gear.

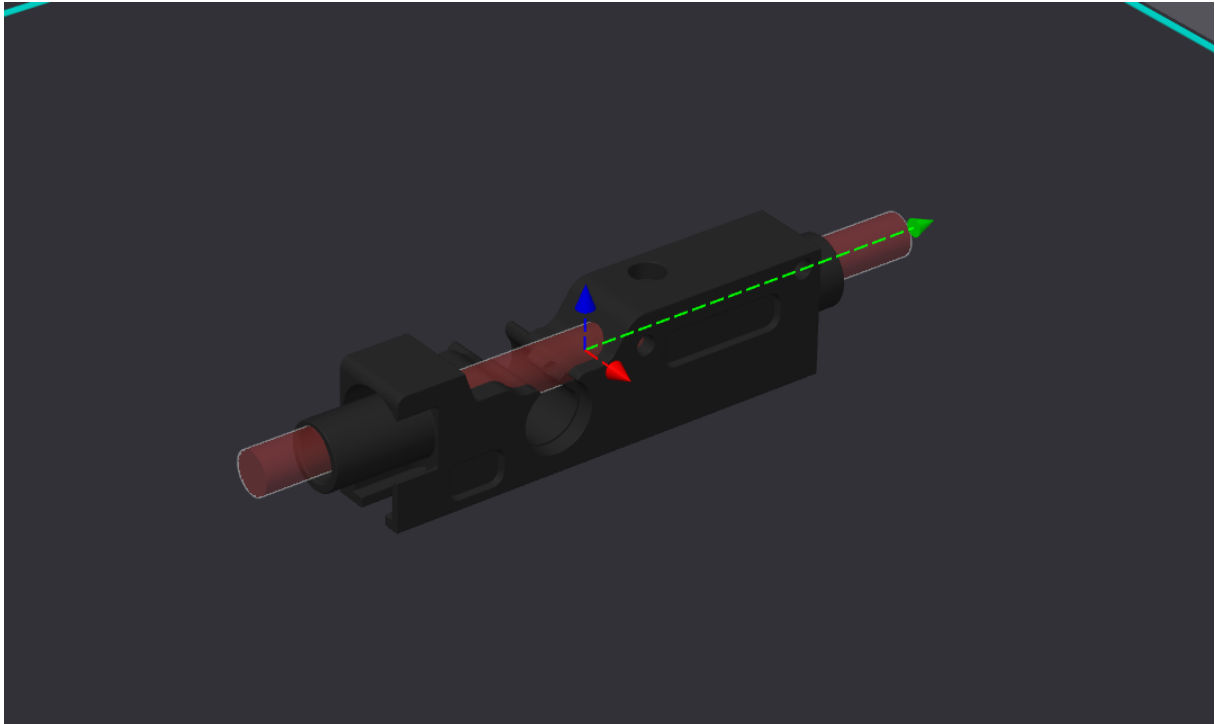
We will release soon a full config and also our members are working in parallel to make how to videos in English and Spanish

As a token of appreciation, this will be available exclusively to our members for the first two weeks, after which it will be open to everyone. Thank you for your understanding and for your continued support of our work!

I have update the Cad and also the T36 The servo Pulley to be aligned with the camshaft

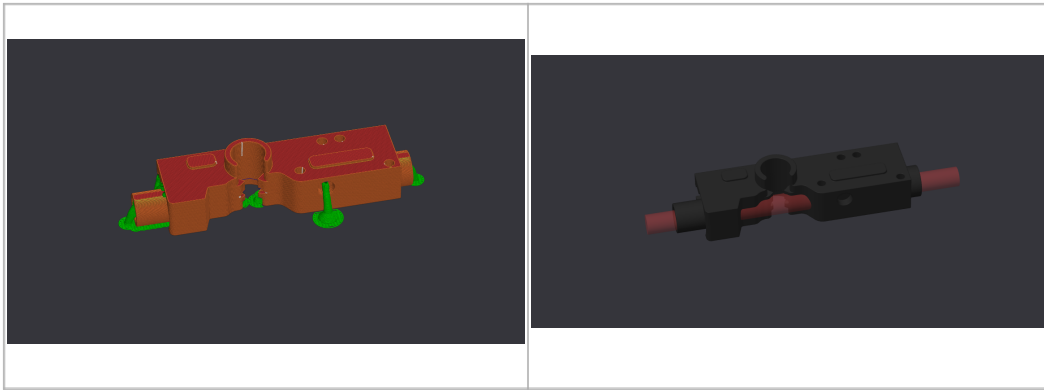
I have also uploaded a PDF with the BOM

I have added a 3mf for the Gate that have a support blocker for the filament path.



this way you will not have support on the filament path that is hard to remove.

if we lay it down on a side you will print it with 5 minute faster



MMX Multi-Material Extruder

Exciting News: MMX Multi-Material Extruder Development Update!

We're thrilled to share that we are making significant progress on our multi-material system! After evaluating the Pico MMU, we've decided to move forward with a **completely redesigned concept** – one that's more user-friendly, customizable, and versatile. ---**Why MMX Stands Out**

Customizable Capacity

- Run not just 4 colors or materials, but as many as you want! Increase or decrease the number of inputs based on your needs. **but for now we stick to 0-1-2-3 Gates**

Reliability Boost

- Equipped with **GT2 closed-loop belts** for enhanced durability.
- Adjustable tensioning ensures no slippage in the system.

Enhanced Filament Handling

- Integrated **Bontech dual gears** for superior filament grip and smoother guidance.

Multi-Functional Use

- Perfect for multi-color or multi-material printing.
- Supports advanced applications such as:

Gate Features

- Each gate features a **pre-gate filament sensor** and **LED indicators**.
- Thanks to the integration with **Happy Hare** (special thanks to Moggie for the help!), the system automates sequences when the pre-gate detects filament in, ensuring smooth operation and many more features.

Hidden Electronics and Sliding Mechanism

- The electronics are neatly **hidden in a box** with a **sliding mechanism** for easy access.

Extruder Design

- The extruder is based on the **Voron V4**, a reliable and powerful extruder with an **80:20 gear ratio**.

Advanced Servo and Camshaft Mechanics

- Gates are engaged by a servo with a **36:20 ratio to the camshaft**, allowing the camshaft to rotate **320 degrees** even when the servo moves only **180 degrees**.
- The camshaft is spaced **90 degrees** for each lobe from the previous one. This results in **50 degrees of servo movement** per operation, thanks to the **1.8:1 ratio** between the servo and the camshaft.

Compatibility

- Designed to work seamlessly with both **direct drive** and **Bowden setups**.
- Compatible with **any Klipper-based 3D printer**. We've made a support mount for the **SV08**, as well as the **Xenomorph**. Additionally, we'll be creating support for the **V4** toolhead, both of which offer a **filament sensor in the toolhead** and a **filament cutter**.

Advanced Hub Options

- Developed a **4-lane hub**.
- Also offering a **4-in-1 hub** featuring an integrated **Klinky encoder**.

Redesigned from Scratch

- This version combines multiple innovative ideas into one cohesive design, with the entire kit **redesigned from scratch** for optimal performance.

Section in Happy Hare

- In Happy Hare, you can find this system under **Section 9: PicoMMU**, with our default cam spacing and design parameters included.

---**Introducing the MMX**The **MMX Multi-Material Extruder** is a cutting-edge solution designed for both hobbyists and professionals. We're

collaborating with **Happy Hare** to ensure smooth integration with their ecosystem. Expect an **easy step-by-step installation guide!** ---

- Using **PETG for PLA** or **ABS for supports**.
- **Infinite Spool Mode:** Load multiple spools of the same material. The system will automatically switch when one spool ends or breaks, eliminating late-night spool monitoring!

Additional Notes:

- **Mounting Compatibility:** Tested with **Xenomorph Orbiter** & **V4** toolheads (both support filament sensor & cutter)
- **Software Recommendation:** **Happy Hare** for seamless MMU integration
- **Beta Status:** Still in beta, improvements ongoing!
- **Upcoming Video:** Assembly & setup guide in progress

Print Settings:

- **Material Recommendations:**
 - PLA or PETG for camshaft & idler doors
 - PETG or ABS for other components
- **Orientation:** Pre-oriented files provided
- **Layer Height:** 0.2mm
- **Walls:** 4
- **TOP/Bottom:** 5
- **Infill:** 15-20
- **Supports:** Minimal (as needed)

Please join the discord for BOM and More!

<https://discord.gg/SJu43FjcuF>

This remix is based on



GitHub - VoronDesign/Mobius-Extruder: Extruder for the Voron line of Printers

Model files



CAD

1 file

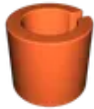


sv08-mmx-alpha-by-cn3d.stp

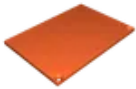
📅 Updated 24/October/2025



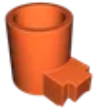
mmx-right-side-cover-spacer-x4.stl



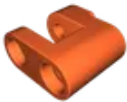
mmx-camshaft-spacer.stl



mmx-back-panel.stl



mmx-led-defuser-x4.stl



mmx-servo-idler.stl



mmx-front-panel.stl



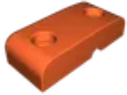
mmx-gate-spacer-x3.stl



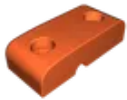
mmx-left-side.stl



mmx-servo-and-extruder-holder.stl



right-ear.stl



left-ear.stl



sv08-mmx-tophat-grile.stl



mmx-camshaft.stl

📄 updated



mmx-t80.stl

📄 4/20



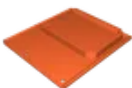
mmx-t36-servo-pulley.stl

📄 updated 16-June-2025



mmx-right-side.stl

📄 updated 16-June-2025



mmx-bottom-panel.stl

📄 Updated 12-jully-2025



mmx-ebb-42-and-stepdown-slide-tray.stl

📄 Updated 12-july-2025



mmx-idler-door-1.stl

📄 Updated 14-july-2025



mmx-idler-door-3.stl

📄 Updated 14-july-2025



mmx-idler-door-0.stl

📄 Updated 14-july-2025



mmx-idler-door-2.stl

📄 Updated 14-july-2025



mmx-gate-x4.stl

📄 updated 24-october-2025



gate-x4.3mf

📄 updated 24-october-2025

Other files



Sovol SV08 Orca Profile

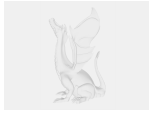
1 file



sovol-sv08-orca-profile-for-mmx.zip



mmx-bom.pdf



mmx-bom-worldwide.pdf

☐ Amazon Affiliate links



mmx-user-manual-24-05-2025.pdf

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